MATH 104 MIDTERM 2 SOLUTION KEY (Spring 2018

Q1) [8pts] **(Curvature and Normal Vectors of a Curve)** Find $\boldsymbol{T}$, $\boldsymbol{N}$ and κ for the space curve

$$\boldsymbol{r}(t) = \cos t\boldsymbol{i} + \sqrt{2}\sin t\boldsymbol{j} + \cos t\boldsymbol{k}.$$

$$\vec{v}(t) = -\sin t\,\vec{i} + \sqrt{2}\cos t\,\vec{j} - \sin t\,\vec{k}$$

$$|\vec{v}(t)| = \sqrt{\sin^2 t + 2\cos^2 t + \sin^2 t} = \sqrt{2}$$

$$\vec{T} = \frac{\vec{v}}{|\vec{v}|} = \frac{1}{\sqrt{2}}\left(-\sin t\,\vec{i} + \sqrt{2}\cos t\,\vec{j} - \sin t\,\vec{k}\right)$$

$$\frac{d\vec{T}}{dt} = \frac{1}{\sqrt{2}}\left(-\cos t\,\vec{i} - \sqrt{2}\sin t\,\vec{j} - \cos t\,\vec{k}\right), \quad \left|\frac{d\vec{T}}{dt}\right| = 1$$

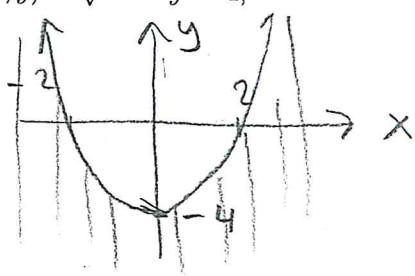
$$\vec{N} = \frac{\frac{d\vec{T}}{dt}}{\left|\frac{d\vec{T}}{dt}\right|} = \frac{1}{\sqrt{2}}\left(-\cos t\,\vec{i} - \sqrt{2}\sin t\,\vec{j} - \cos t\,\vec{k}\right)$$

$$K = \frac{1}{|\vec{v}|}\left|\frac{d\vec{T}}{dt}\right| = \frac{1}{\sqrt{2}}\,1 = \frac{1}{\sqrt{2}}$$

Q2) [2×4 =8pts] **(Functions of Several Variables)** For the function $f(x,y) = \sqrt{x^2 - y - 4}$,

(i) Find and sketch the domain of f.

D_f: $x^2 - y - 4 \geq 0 \Rightarrow y \leq x^2 - 4$



(ii) Determine if the domain is an open region, a closed region or neither.

D_f is a closed region

(iii) Decide if the domain is bounded or unbounded.

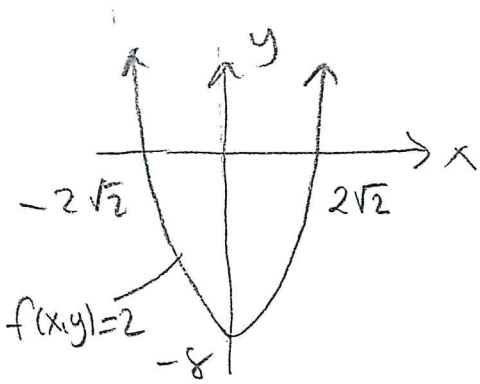
D_f is an unbounded region

(iv) Find and sketch the level curve $f(x,y) = 2$.

$$f(x,y) = 2 \Rightarrow \sqrt{x^2 - y - 4} = 2$$

$$\Rightarrow x^2 - y - 4 = 4$$

$$\Rightarrow y = x^2 - 8$$



Q3) [5+5=10pts] **(Limits and Continuity in Higher Dimensions)** Find each limit or show that the limit does not exist.

(i) $\lim_{\substack{(x,y)\to(1,1)\\ x\neq y}} \frac{x^2-y^2}{x-y} = \lim_{\substack{(x,y)\to(1,1)\\ x\neq y}} \frac{\cancel{(x-y)}(x+y)}{\cancel{x-y}} = 2$

(ii) $\lim_{\substack{(x,y)\to(0,0)\\ xy\neq 0}} \frac{x^2+y^2}{xy} = \lim_{\substack{x\to 0\\ y=kx}} \frac{\cancel{x^2}(1+k^2)}{\cancel{x^2}k} = \frac{1+k^2}{k}$ different along different paths

Let $y = kx$, $k \in \mathbb{R}$.

By the two-path test, the limit does not exist

Q4) [8pts] **(Partial Derivatives)** Show that the function $f(x,y,z) = e^{3x+4y}\cos(5z)$ satisfies the Laplace equation

$$f_{xx} + f_{yy} + f_{zz} = 0.$$

$f_x = 3e^{3x+4y}\cos(5z)$

$f_y = 4e^{3x+4y}\cos(5z)$

$f_z = -5e^{3x+4y}\sin(5z)$

$f_{xx} = 9e^{3x+4y}\cos(5z) = 9f$

$f_{yy} = 16e^{3x+4y}\cos(5z) = 16f$

$+ \quad f_{zz} = -25e^{3x+4y}\cos(5z) = -25f$

$f_{xx} + f_{yy} + f_{zz} = (9+16-25)f = 0$

Q5) [2+6=8pts] **(The Chain Rule)** *(i)* Write a Chain Rule formula for $\frac{dw}{dt}$ if $w = f(x, y)$, $x = x(t)$ and $y = y(t)$.

$$\frac{dw}{dt} = \frac{\partial w}{\partial x}\frac{dx}{dt} + \frac{\partial w}{\partial y}\frac{dy}{dt}$$

(ii) Find $\frac{dw}{dt}$ at $t = 0$ if $\frac{\partial w}{\partial x} = y\cos(xy + \pi)$, $\frac{\partial w}{\partial y} = x\cos(xy + \pi)$, $x = e^t$ and $y = \ln(t + 1)$.

For $t=0$, $x=e^0=1$ and $y=\ln(0+1)=0$

$$\frac{dw}{dx} = y\cos(xy+\pi)\, e^t + x\cos(xy+\pi)\,\frac{1}{t+1}$$

$$\left.\frac{dw}{dt}\right|_{t=0} = 0 \cdot \cos\pi\, e^0 + 1 \cdot \cos\pi \;\; 1 = -1$$

Q6) [8pts] **(The Chain Rule)** Find the values of $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$ at the point $(1, 1, 1)$ if $\ln(xyz) - x^2 - 2xyz^2 + 3 = 0$.

$F(x,y,z) = \ln(xyz) - x^2 - 2xyz^2 + 3 = 0 \Rightarrow z = f(x,y)$ implicitly.

$F_x = \frac{1}{x} - 2x - 2yz^2$ $\quad F_y = \frac{1}{y} - 2xz^2$ $\quad F_z = \frac{1}{z} - 4xyz$

$$\frac{\partial z}{\partial x} = -\frac{F_x}{F_z} = -\frac{\frac{1}{x} - 2x - 2yz^2}{\frac{1}{z} - 4xyz} \Rightarrow \left.\frac{\partial z}{\partial x}\right|_{(1,1,1)} = -\frac{1-2-2}{-3} = -1$$

$$\frac{\partial z}{\partial y} = -\frac{F_y}{F_z} = -\frac{\frac{1}{y} - 2xz^2}{\frac{1}{z} - 4xyz} \Rightarrow \left.\frac{\partial z}{\partial y}\right|_{(1,1,1)} = -\frac{1-2}{-3} = -\frac{1}{3}$$

Q7) [3+3+2=8pts] **(Directional Derivatives and Gradient Vectors)**

Let $f(x, y, z) = x^2 + xy - 2y^3z - z^2$.

(i) Find ∇f at the point $(2, -1, 1)$.

$$\vec{\nabla} f = f_x \vec{i} + f_y \vec{j} + f_z \vec{k} = (2x+y)\vec{i} + (x - 6y^2 z)\vec{j} + (-2y^3 - 2z)\vec{k}$$

$$\vec{\nabla} f(2,-1,1) = (4-1)\vec{i} + (2-6)\vec{j} + (2-2)\vec{k} = 3\vec{i} - 4\vec{j}$$

(ii) Find the derivative of f at the point $P(2, -1, 1)$ in the direction of $\boldsymbol{v} = \boldsymbol{i} + 2\boldsymbol{j} - 2\boldsymbol{k}$.

not a unit vector

$$\vec{u} = \frac{\vec{v}}{|\vec{v}|} = \frac{1}{\sqrt{1+4+4}}(\vec{i} + 2\vec{j} - 2\vec{k}) = \frac{1}{3}(\vec{i} + 2\vec{j} - 2\vec{k})$$

$$(D_{\vec{u}} f)_P = \vec{\nabla} f(P) \cdot \vec{u} = (3\vec{i} - 4\vec{j}) \cdot \frac{1}{3}(\vec{i} + 2\vec{j} - 2\vec{k}) = \frac{1}{3}(3-8) = \frac{-5}{3}$$

(iii) Find the direction $\boldsymbol{u}$ and the value of $D_{\boldsymbol{u}} f(2, -1, 1)$ for which $D_{\boldsymbol{u}} f(2, -1, 1)$ is <u>smallest</u>.

Direction v: $\vec{u} = -\dfrac{\vec{\nabla} f(P)}{|\vec{\nabla} f(P)|} = -\dfrac{(3\vec{i} - 4\vec{j})}{\sqrt{9+16}} = \dfrac{-3\vec{i} + 4\vec{j}}{5}$

Smallest value: $(D_{\vec{u}} f)_P = -|\vec{\nabla} f(P)| = -5$.

Q8) [6pts] **(Tangent Planes and Differentials)** Find an equation for the plane tangent to the surface $z = \ln(x^2 + y^2)$ at the point $P(0, e, 2)$.

Let $f(x,y,z) = \ln(x^2+y^2) - z = 0$

Eq. for the line tangent to the surface $f(x,y,z) = 0$ is

$$f_x(P)(x - x_0) + f_y(P)(y - y_0) + f_z(P)(z - z_0) = 0$$

$f_x = \dfrac{2x}{x^2+y^2}$, $f_y = \dfrac{2y}{x^2+y^2}$, $f_z = -1$; $f_x(P) = 0$ $f_y(P) = \frac{2}{e}$, $f_z(P) = -1$

$$0(x-0) + \frac{2}{e}(y - \cancel{e}) - (z - \cancel{2}) = 0 \Rightarrow z = \frac{2}{e}y$$

Q9) [10pts] **(Extreme Values and Saddle Points)** Find all local maxima, local minima and saddle points of the function $f(x, y) = x^3 + y^3 - 3xy + 15$.

$f_x = 3x^2 - 3y$ $\qquad$ $f_y = 3y^2 - 3x$

$f_{xx} = 6x$ $\qquad$ $f_{yy} = 6y$ $\qquad$ $f_{xy} = -3$

CPs: $\left.\begin{array}{l} f_x = 0 \Rightarrow y = x^2 \\ f_y = 0 \Rightarrow x = y^2 \end{array}\right\}$ $y = y^4 \Rightarrow y(1-y^3) = 0 \Rightarrow \begin{array}{l} y = 0, \ x = 0 \\ y = 1, \ x^2 = 1 \end{array}$

$x = 1$, $\cancel{x = -1}$ since $x = y^2$

CPS $(0,0)$ $(1,1)$

$(0,0)$: $D(0,0) = f_{xx}(CP)\, f_{yy}(CP) - f_{xy}^2(CP) = -9 < 0$ ie $(0,0)$ is a saddle point

$(1,1)$: $D(1,1) = f_{xx}(CP)\, f_{yy}(CP) - f_{xy}^2(CP) = 36 - 9 > 0$ anf $f_{xx}(CP) = 6 > 0$

By the Second Derivative Test, f has a local min at $(1,1)$

Q10) [8pts] **(Lagrange Multipliers)** Use the method of Lagrange Multipliers to find the maximum and minimum values of $f(x, y) = 2\sqrt{3}x - 2y + 6$ on the circle $x^2 + y^2 = 4$.

$\left.\begin{array}{l} \vec{\nabla} f = \lambda \vec{\nabla} g \\ g = 0 \end{array}\right\}$ $\qquad$ $2\sqrt{3}\vec{i} - 2\vec{j} = \lambda(2x\vec{i} + 2y\vec{j})$

① $\cancel{2}\sqrt{3} = \lambda \cancel{2} x$ $\qquad$ ② $-\cancel{2} = \lambda \cancel{2} y$ $\qquad$ ③ $x^2 + y^2 = 4$

$x = \frac{\sqrt{3}}{\lambda}$ $\qquad$ $y = -\frac{1}{\lambda}$ $\qquad$ $\frac{3}{\lambda^2} + \frac{1}{\lambda^2} = 4$

$\lambda^2 = 1$

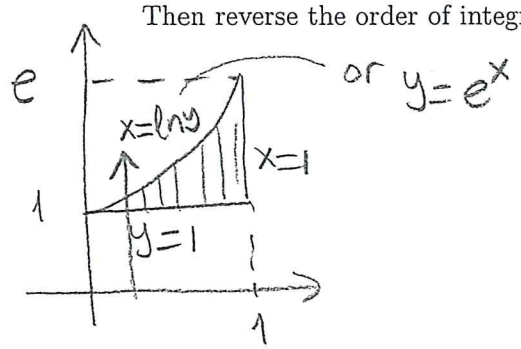
$\lambda = 1$: $(\sqrt{3}, -1)$ $\qquad$ $\lambda = -1$: $(-\sqrt{3}, 1)$

$f(\sqrt{3}, -1) = 6 + 2 + 6 = 14$ max value $\qquad$ $f(-\sqrt{3}, 1) = -6 - 2 + 6 = 2$ min. value

Q11) [10pts] **(Double Integrals over General Regions)** Sketch the integration region for the following integral

$$\int_1^e \int_{\ln y}^1 y\,dx\,dy.$$

Then reverse the order of integration and evaluate the integral.



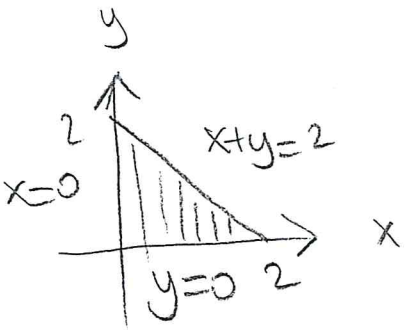
$$I=\int_{x=0}^{1}\int_{y=1}^{e^x} y\,dy\,dx$$

$$=\int_0^1 \left[\frac{y^2}{2}\right]_1^{e^x} dx = \frac{1}{2}\int_0^1 (e^{2x}-1)\,dx$$

$$=\frac{1}{2}\left(\frac{1}{2}e^{2x}-x\right)_0^1 = \frac{1}{2}\left(\frac{e^2}{2}-1-\frac{1}{2}\right)=\frac{1}{4}(e^2-3)$$

Q12) [8pts] **(Double Integrals over General Regions)** Find the volume of the solid bounded above by the plane $z = 2 + x$ and below by the triangular region bounded by the lines $x = 0,\ y = 0$ and $x + y = 2$.

$$V=\iint_R z(x,y)\,dA = \iint_R (2+x)\,dA$$



$$V=\int_0^2\int_0^{2-x}(2+x)\,dy\,dx$$

$$V=\int_0^2 (2+x)[y]_0^{2-x}\,dx$$

$$V=\int_0^2 (2+x)(2-x)\,dx = \int_0^2 (4-x^2)\,dx = \left[4x-\frac{x^3}{3}\right]_0^2 = \frac{16}{3}$$